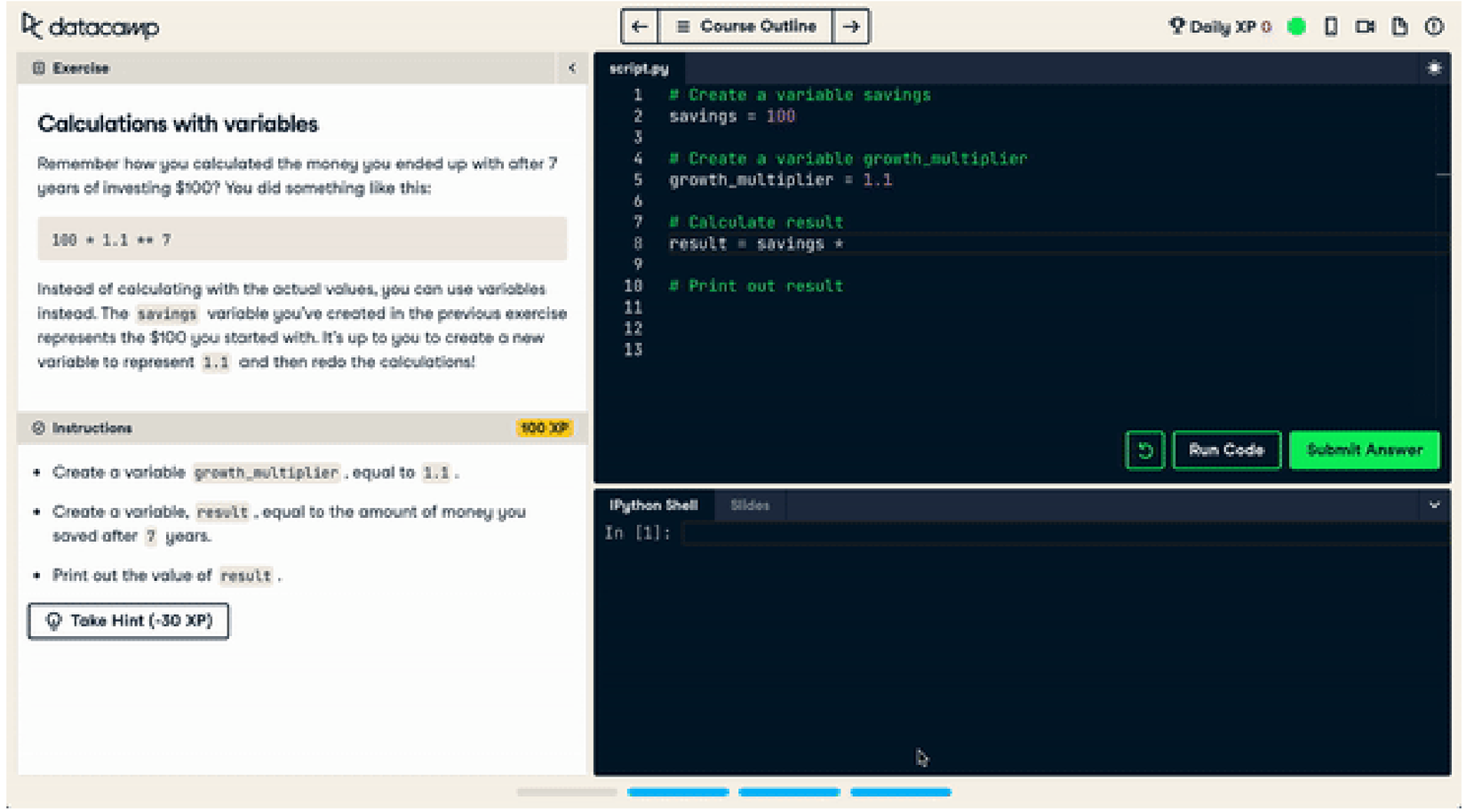
Hello Python!

INTRODUCTION TO PYTHON

Hugo Bowne-Anderson
Data Scientist at DataCamp

datacamp

How you will learn



Python

- General purpose: build anything
- Open source! Free!
- Python packages, also for data science
  - Many applications and fields
- Version 3.x - **https://www.python.org/downloads/**

IPython Shell

Execute Python commands

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Course Outline

Daily XP 100

Exercise

Calculations with variables

Remember how you calculated the money you ended up with after 7 years of investing $100? You did something like this:

```
100 * 1.1 ** 7
```

Instead of calculating with the actual values, you can use variables instead. The `savings` variable you've created in the previous exercise represents the $100 you started with. It's up to you to create a new variable to represent `1.1` and then redo the calculations!

Instructions 100 XP

- Create a variable `growth_multiplier`, equal to `1.1`.
- Create a variable, `result`, equal to the amount of money you saved after `7` years.
- Print out the value of `result`.

Take Hint (-30 XP)

script.py

```
# Create a variable savings
savings = 100

# Create a variable growth_multiplier
growth_multiplier = 1.1

# Calculate result
result = savings * growth_multiplier ** 7

# Print out result
print(result)
```

Run Code | Submit Answer

IPython Shell | Slides

```
In [1]:
```

IPython Shell

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IPython Shell Slides

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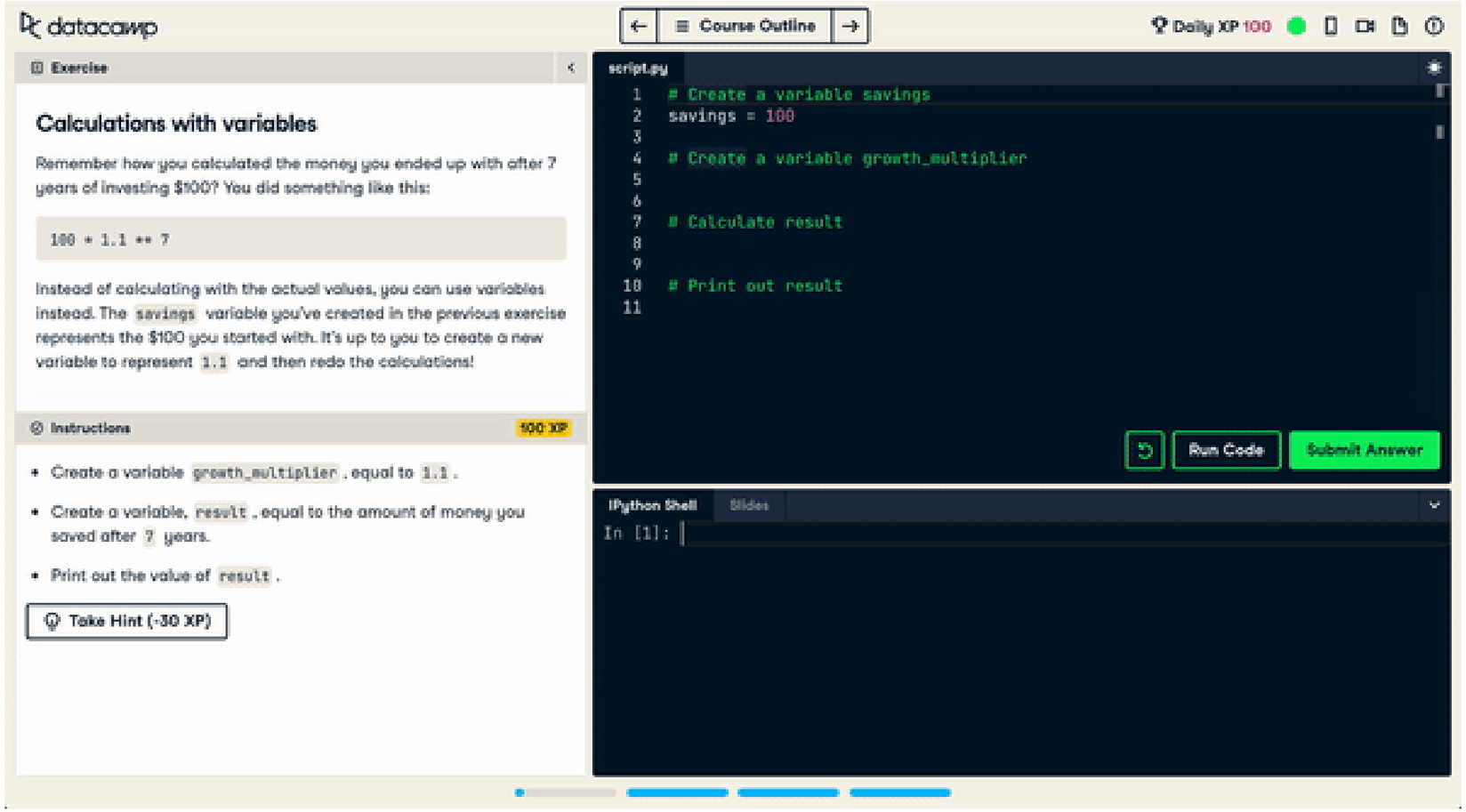
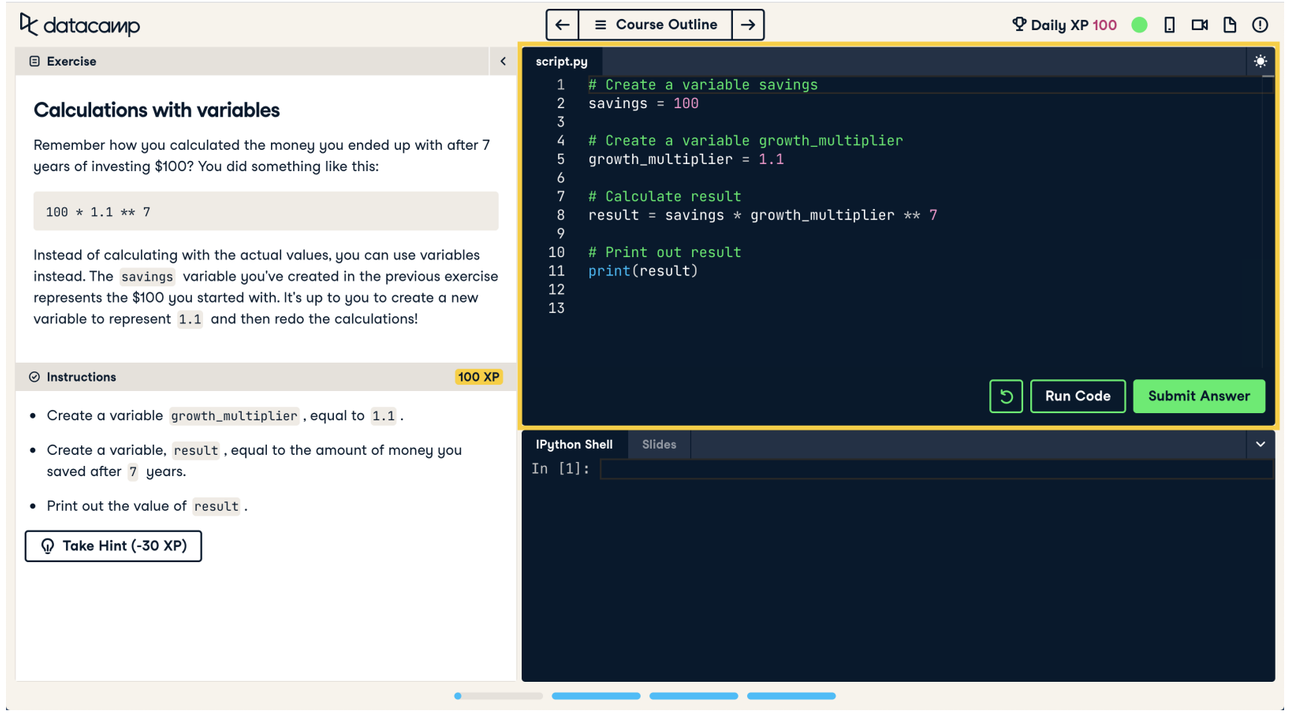
IPython Shell

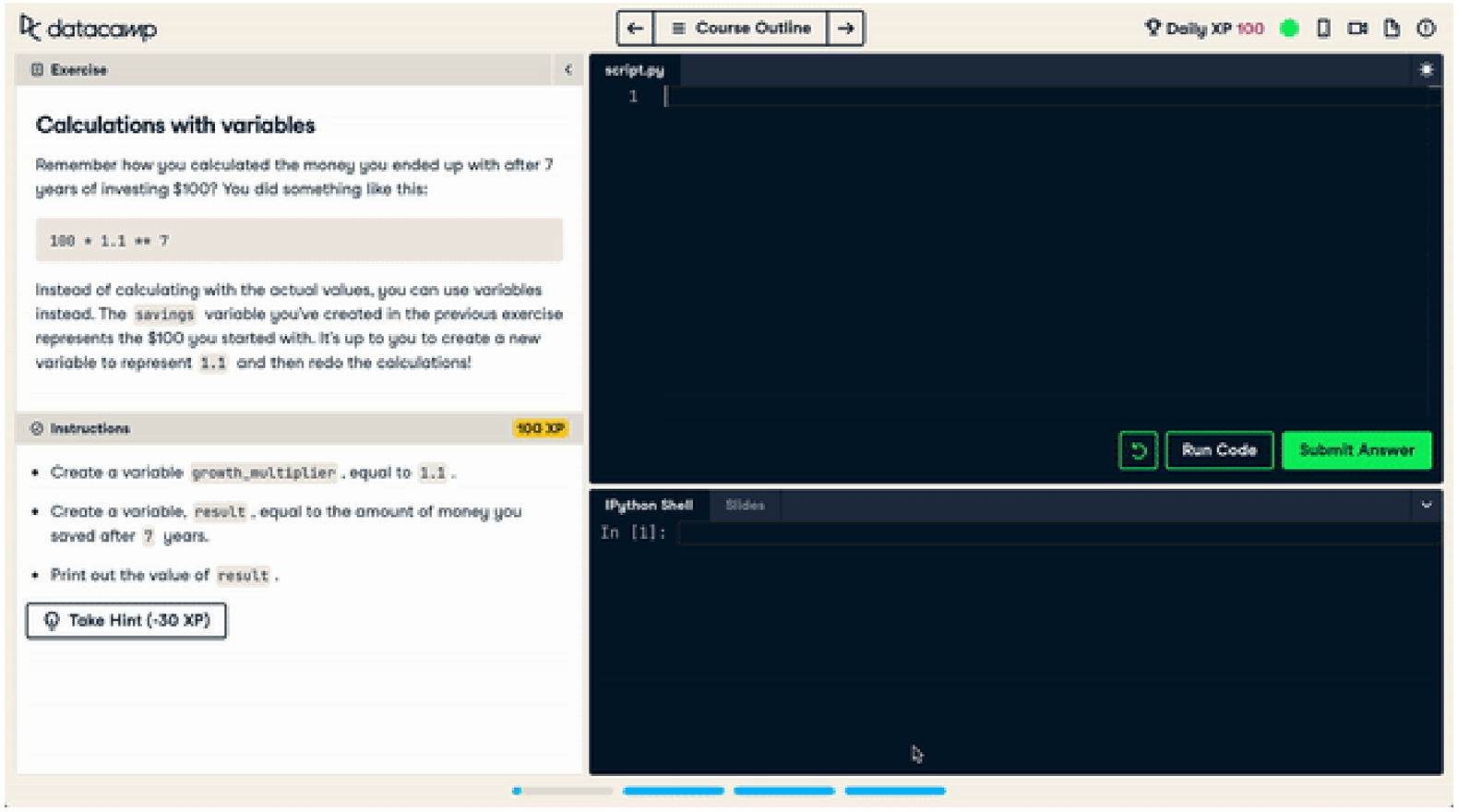


Python Script

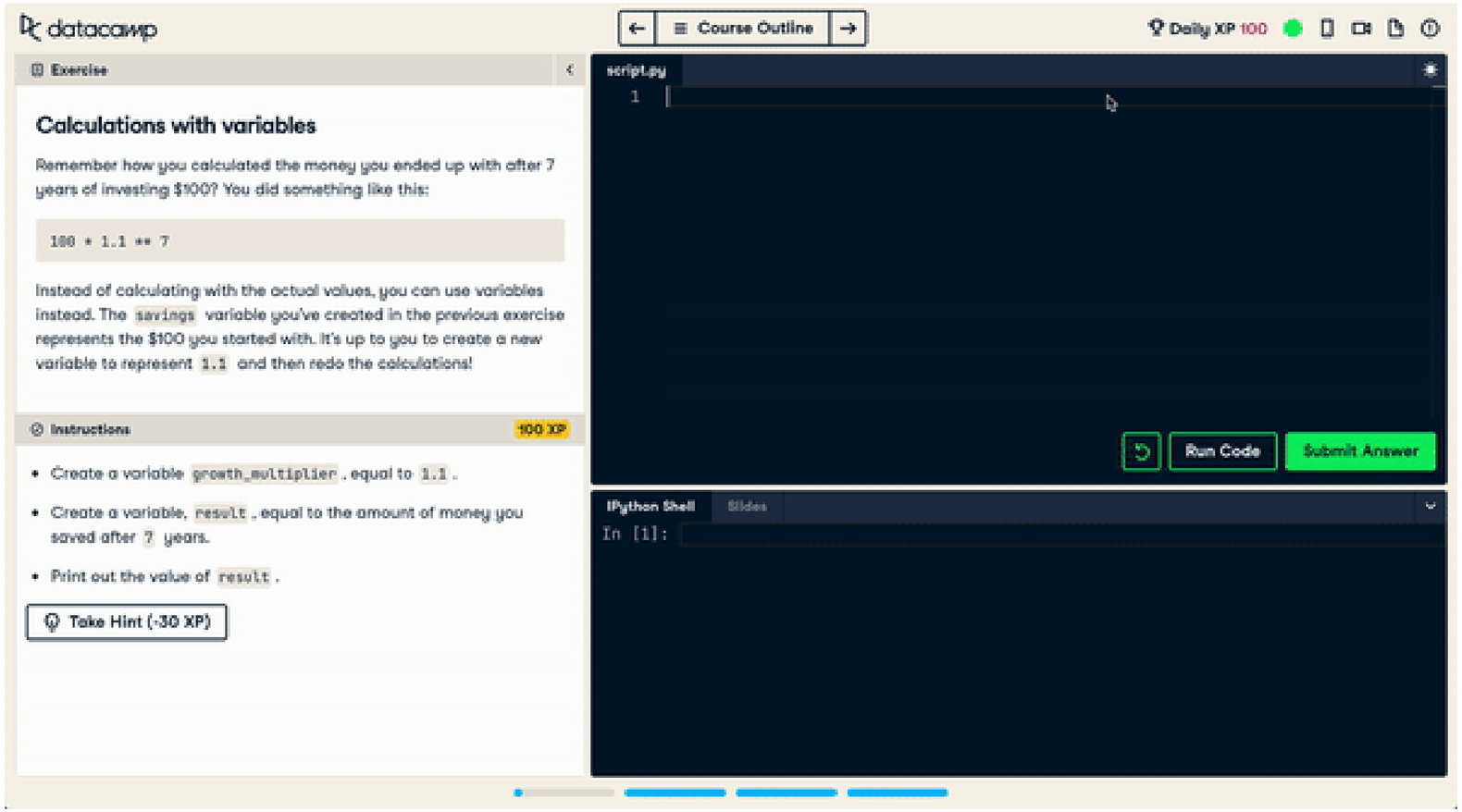
- Text files - `.py`
- List of Python commands
- Similar to typing in IPython Shell



Python Script



Python Script



- Use `print()` to generate output from script

DataCamp Interface

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Run Code

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IPython Shell Slides

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In [1]:
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Let's practice!

INTRODUCTION TO PYTHON

Variables and Types

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Variable

- Specific, case-sensitive name
- Call up value through variable name
- 1.79 m - 68.7 kg

```
height = 1.79
weight = 68.7
height
```

```
1.79
```

Calculate BMI

```
height = 1.79
weight = 68.7
height
```

```
1.79
```

$$\text{BMI} = \frac{\text{weight}}{\text{height}^2}$$

```
68.7 / 1.79 ** 2
```

```
21.4413
```

```
weight / height ** 2
```

```
21.4413
```

```
bmi = weight / height ** 2
bmi
```

```
21.4413
```

Reproducibility

```
height = 1.79
weight = 68.7
bmi = weight / height ** 2
print(bmi)
```

```
21.4413
```

Reproducibility

```
height = 1.79
weight = 74.2 # <-
bmi = weight / height ** 2
print(bmi)
```

```
23.1578
```

Python Types

```
type(bmi)
```

```
float
```

```
day_of_week = 5
type(day_of_week)
```

```
int
```

Python Types (2)

```
x = "body mass index"
y = 'this works too'
type(y)
```

```
str
```

```
z = True
type(z)
```

```
bool
```

Python Types (3)

```
2 + 3
```

```
5
```

```
'ab' + 'cd'
```

```
'abcd'
```

- Different type = different behavior!

Let's practice!

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